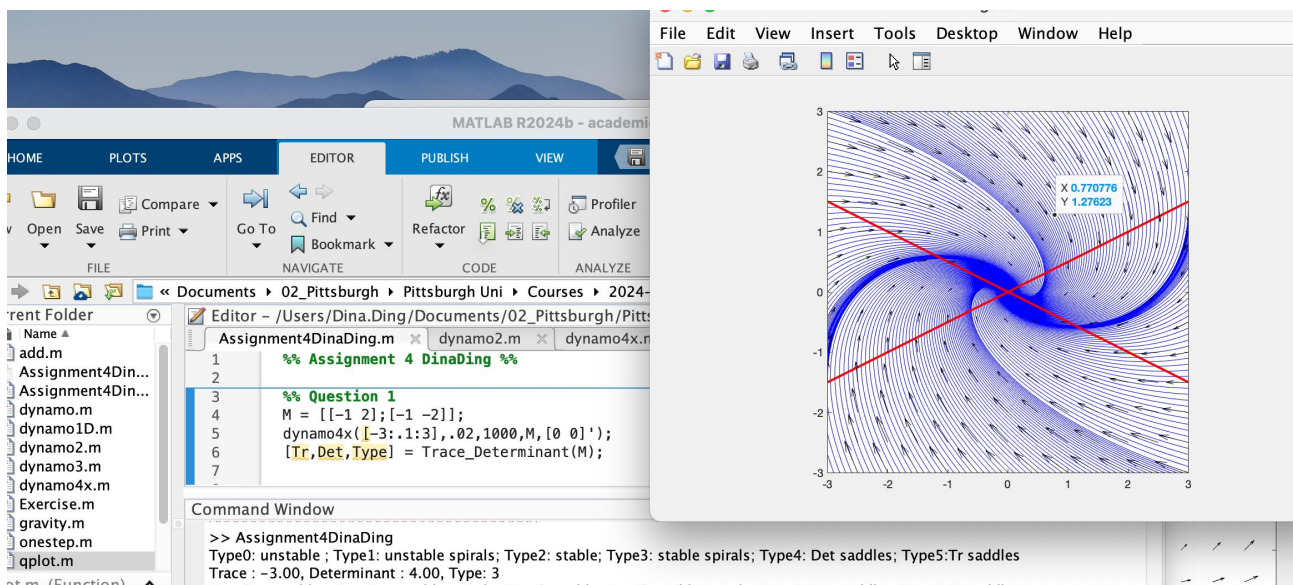


## Assignment 4 / Dina Ding / Mon Feb,03,2025

1. Simulate the linear dynamical system using **dynamo4x.m** (slides 35 and 36) given by the matrix  $\mathbf{M} = \begin{bmatrix} -1 & 2 \\ -1 & -2 \end{bmatrix}$ . How would you describe the equilibrium point? Compute the Trace and Determinant. Check if it consistent with the dynamics as plotted.

Using the functions **dynamo4x.m** and **Trace\_Determinant.m** to plot and calculate the value of Trace and Determinant.

The plot is showed below, the Trace = -3; Determinant = 4, and it is Type3- Stable Spiral and its equilibrium point is (0,0) . This calculated result is consistent with the plot. This means all trajectories spiral toward the equilibrium over time.



```
Editor - /Users/Dina.Ding/Documents/02_Pittsburgh/Pittsburgh Uni/Courses/2024-Spring
Assignment4DinaDing.m x dynamo2.m x dynamo4x.m x qplot.m x Trace_Determinant.m

1 function ptseq = dynamo4x( rang,step,niter,cmtx,cvct )
2 figure
3 xinit=[rang rang rang(1)*ones(size(rang)) rang(end)*ones(size(rang))];
4 yinit=[rang(1)*ones(size(rang)) rang(end)*ones(size(rang)) rang rang];
5 [xcr,ycr] = meshgrid(linspace(rang(1),rang(end),100));
6 nullx=cmtx(1,1)*xcr+cmtx(1,2)*ycr+cvct(1,1);
7 nullx=cmtx(2,1)*xcr+cmtx(2,2)*ycr+cvct(2,1);
8 % size(xinit)
9 for k=1:max(size(xinit))
10 ptseq = [xinit(k);yinit(k)];
11 curr = [xinit(k);yinit(k)];
12 for i=1:niter
13 curr=onestep(curr,cmtx,cvct,step);
14 ptseq=[ptseq,curr];
15 end
16 plot(ptseq(1,:),ptseq(2:,:), 'blue');
17 axis equal;
18 hold on;
19 end
20 axis([rang(1) rang(end) rang(1) rang(end)]);
21 contour(xcr,ycr,nullx,[0 0], 'red', 'LineWidth',2) ;
22 contour(xcr,ycr,nullx,[0 0], 'red', 'LineWidth',2) ;
23 qplot(cmtx,cvct,rang(1):.5:rang(end))
24 end
25
```

```

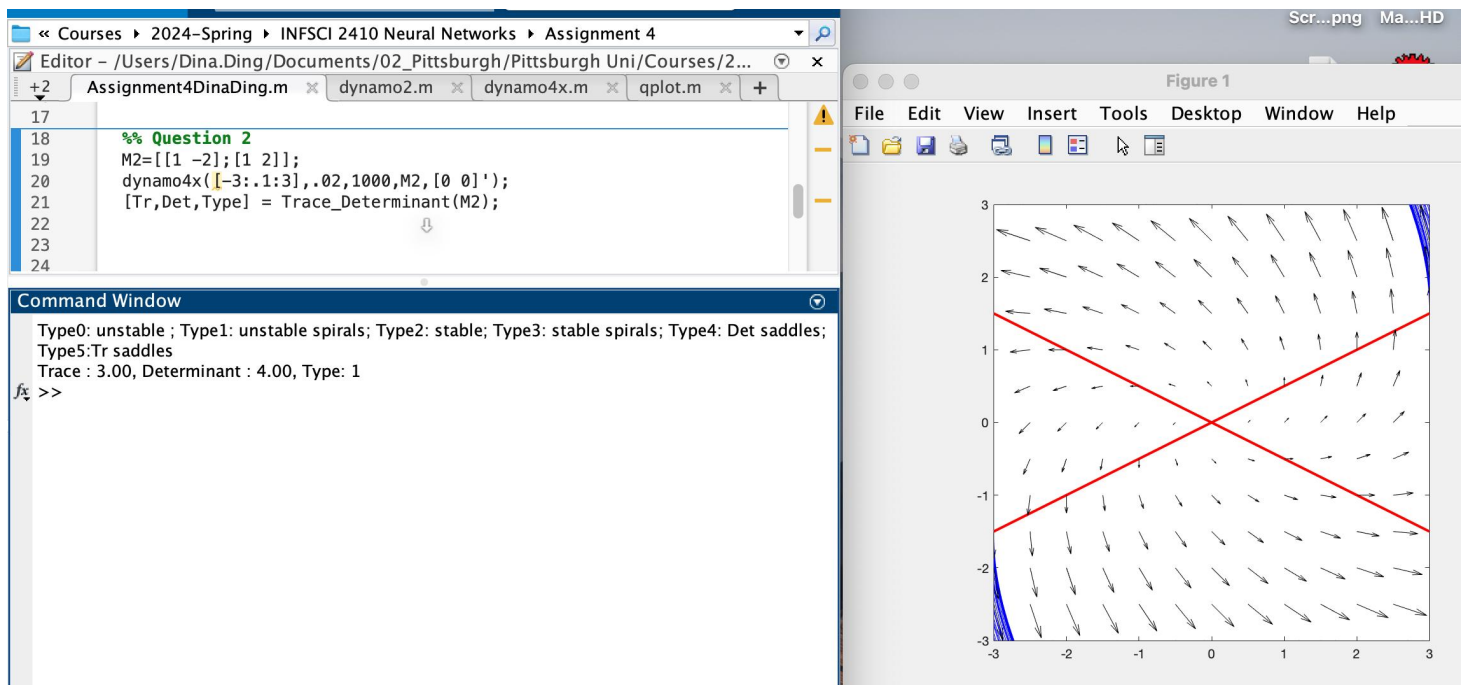
1  %% Trace and Determinant
2  function [Tr Det Type] = Trace_Determinant(M)
3  Tr=M(1,1)+M(2,2);
4  Det=M(1,1)*M(2,2)-M(1,2)*M(2,1);
5  if Tr > 0
6      if Tr^2>4*Det
7          Type = 0; % unstable
8      else
9          Type = 1; % unstable spirals
10     end
11 elseif Tr < 0
12     if Tr^2>4*Det
13         Type = 2; % stable
14     else
15         Type = 3; % stable spirals
16     end
17 elseif Tr == 0
18     if Det >0
19         Type =4; % Det saddles
20     else
21         Type =5; % Tr saddles
22     end
23 end
24
25 fprintf('Type0: unstable ; Type1: unstable spirals; Type2: stable; Type3: stable spirals; Type4: Det saddles; Type5:Tr saddles\n');
26 fprintf('Trace : %.2f, Determinant : %.2f, Type: %d\n',Tr,Det,Type);
27 end

```

2. Repeat by using a new matrix that is  $-M$ . How is it different?

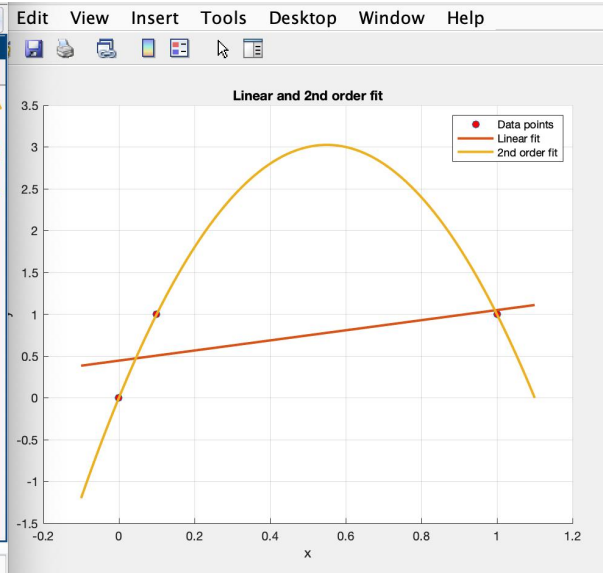
If we use  $-M$ , the plot is different.

The plot is showed below, the Trace = 3 (different from M which is -3); Determinant =4, and it is Type change to 1: Unstable Spiral , its equilibrium point is still (0,0) . The calculated result of Tr & Det is consistent with the plot. This means the system is unstable.



3. Fit a second order polynomial to the data points (0,0), (0.1,1), and (1,1). Plot the result and compare this to a linear fit (1st order polynomial).

```
<< Courses > 2024-Spring > INFSCI 2410 Neural Networks > Assignment 4
Editor - /Users/Dina.Ding/Documents/02_Pittsburgh/Pittsburgh Uni/Courses/2...
Assignment4DinaDing.m x dynamo2.m x dynamo4x.m x qplot.m x +
14
15 %% Question 3
16 x=[0,.1,1];
17 y=[0,1,1];
18 linear_fit=polyfit(x,y,1);
19 second_fit=polyfit(x,y,2);
20 x_fit=linspace(-.1,1.1,100); % generate points for plotting the fi
21 y_linear=polyval(linear_fit,x_fit);
22 y_second=polyval(second_fit,x_fit);
23
24 figure;
25 hold on;
26 scatter(x,y,'MarkerFaceColor','r');
27 plot(x_fit,y_linear,"linewidth",2);
28 plot(x_fit,y_second,"LineWidth",2);
29 legend('Data points','Linear fit','2nd order fit');
30 xlabel('x');
31 ylabel('y');
32 title('Linear and 2nd order fit');
33 grid on;
34 hold off;
35
```



Command Window